Mathematical Analysis Methods

Professor Shunming Zhang

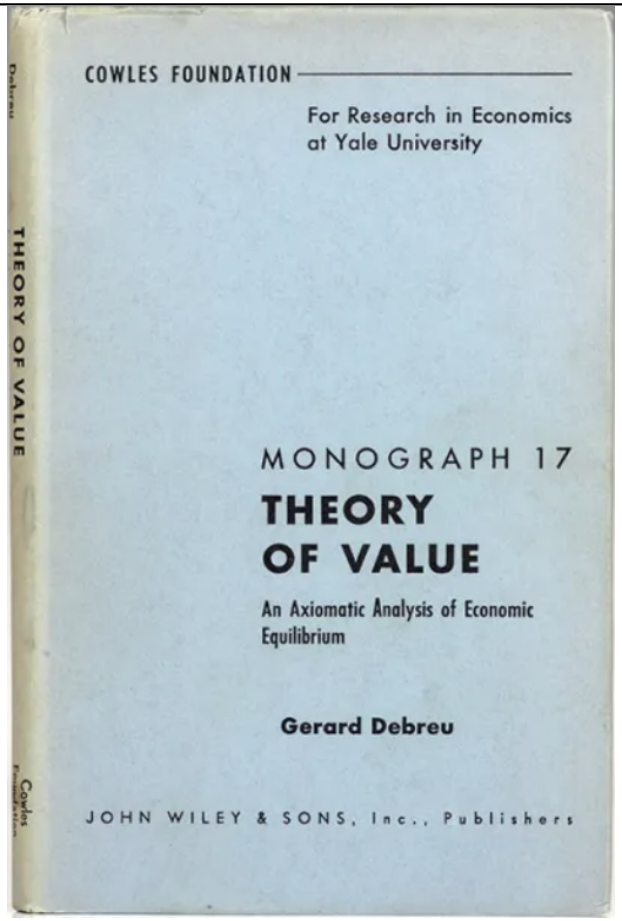
School of Finance
Renmin University of China

September 10, 2022

Mathematical Analysis Methods

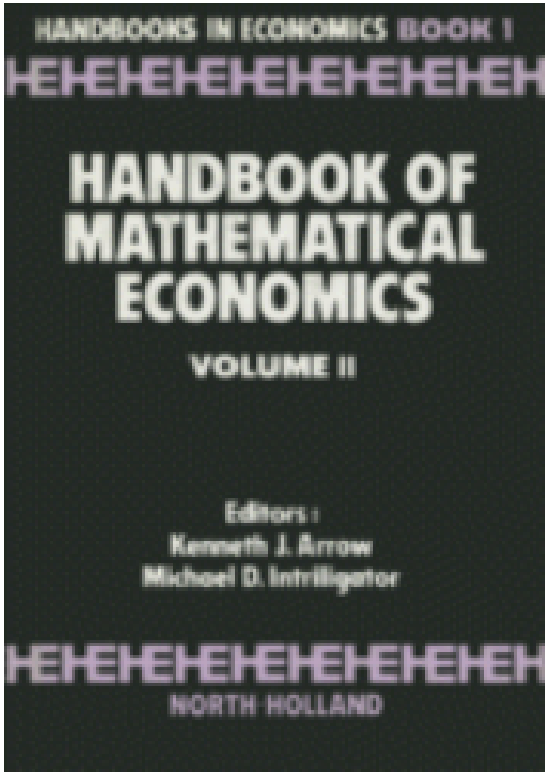
1. Gerard Debreu (1959), **Theory of Value — An Axiomatic Analysis of Economic Equilibrium**
2. Gerard Debreu (1982), *Existence of Competitive Equilibrium*. In Kenneth J. Arrow and Michael D. Intriligator (1982), **Handbook of Mathematical Economics (Volume 2)**, Chapter 15, Pages 697 - 743.
3. Gerard Debreu (1983), **Mathematical Economics** — Twenty Papers of Gerard Debreu.
4. Claude Berge (1959), **Espaces Topologiques**. Translated as **Topological Spaces** (1963)

Gerard Debreu (1959)



1. *Mathematics*
2. *Commodities and Prices*
3. *Producers*
4. *Consumers*
5. *Equilibrium*
6. *Optimum*
7. *Uncertainty*

Gerard Debreu (1982)



Chapter 15

EXISTENCE OF COMPETITIVE EQUILIBRIUM*

GERARD DEBREU

University of California

1. Introduction

The mathematical model of a competitive economy of L. Walras (1874–77) was conceived as an attempt to explain the state of equilibrium reached by a large number of small agents interacting through markets. Walras himself perceived that the theory that he proposed would be vacuous without a mathematical argument in support of the existence of at least one equilibrium state. However, for more than half a century the equality of the number of equations and of the number of unknowns of the Walrasian model remained the only, unconvincing, remark made in favor of the existence of a competitive equilibrium. Study of the existence problem began in the early 1930s when Neisser (1932), Stackelberg (1933), Zeuthen (1933), and Schlesinger (1935) identified several of its basic features and when Wald (1935, 1936a, 1936b) obtained its first solution. After an interruption of some twenty years, the question of existence of an economic equilibrium was taken up again by Arrow and Debreu (1954), McKenzie (1954, 1955), Gale (1955), Debreu (1956), Kuhn (1956a, 1956b), Nikaido (1956), Uzawa (1956), and many other authors whose contributions over the last twenty-five years constitute a bibliography of over three hundred and fifty items.

In this literature four distinct, but closely related, approaches to the existence problem can be recognized. (1) At first, proofs of existence of an economic equilibrium were uniformly obtained by application of a fixed-point theorem of the Brouwer type or of the Kakutani type or by analogous arguments. This approach, which has remained of central importance to the present, is the subject of this chapter. (2) In the last decade, efficient algorithms of a combinatorial nature for the computation of an approximate economic equilibrium were developed. These algorithms, which provide a constructive answer to the existence

*More than forty persons have made suggestions that were incorporated in the text or in the bibliography of this chapter. I wish to express my thanks to each one of them. The support of the National Science Foundation in the preparation of this article is also gratefully acknowledged.

Handbook of Mathematical Economics, vol. II, edited by K.J. Arrow and M.D. Intriligator
© North-Holland Publishing Company, 1982

Theorem 5

The economy $\mathcal{E}$ has an equilibrium if for every i,
X_i is closed, convex, and has a lower bound for $\leqq$,
there is no satiation consumption in $\hat{X}_i$,
the set $\{(x, x') \in X_i \times X_i | x \precsim_i x'\}$ is closed,
if x and x' are two points of X_i such that $x \prec_i x'$, and r is a real number in $]0, 1]$, then $x \prec_i (1-r)x + rx'$,
there is x_i^0 in X_i such that $x_i^0 \ll e_i$;
for every j,
$0 \in Y_j$;
Y is closed and convex,
$Y \cap (-Y) = \{0\}$,
$Y \supset (-R_+^\ell)$.

The lower boundedness assumption on the consumption sets is easily justified since the quantity of any commodity that an individual actually consumes is bounded below by zero, while the quantity of any type of labor that he produces during a certain time period is obviously bounded above in absolute value.

The convexity of the total production set Y expresses the idea that the aggregate technology of the economy obeys the law of non-increasing returns to scale. The irreversibility assumption $Y \cap (-Y) = \{0\}$ means that if a total production y different from 0 is possible, then the production $-y$ (in which all the inputs have become outputs and vice versa) is not possible. The last assumption explicitly postulates free disposal of all commodities.

Proof of Theorem 5

The proof consists of constructing an economy $\hat{\mathcal{E}}$ satisfying the assumptions of Theorem 4 and such that a free disposal-equilibrium of $\hat{\mathcal{E}}$ yields an equilibrium of $\mathcal{E}$.

In the first step of this construction we replace in the definition of $\mathcal{E}$, each production set Y_j by the closure $\bar{Y}_j$ of its convex hull $\hat{Y}_j$, remarking that

$$\sum_{j=1}^{n} \bar{Y}_j = Y.$$

The proof of this equality rests on two facts:

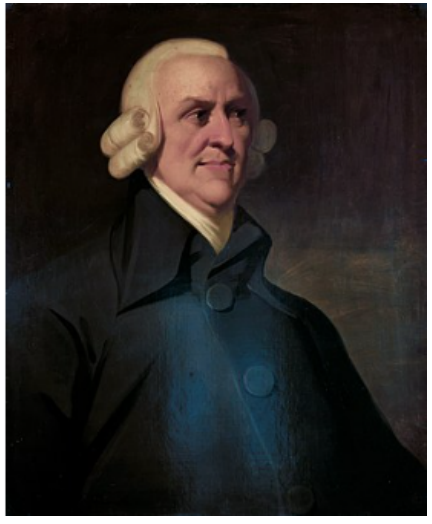
(1) The sum of the convex hulls of a finite collection of sets is equal to the convex hull of their sum. Since $\sum_{j=1}^n Y_j = Y$ has been assumed to be convex, we have $\sum_{j=1}^n \bar{Y}_j = Y$.
(2) The sum of the closures of a finite collection of sets is contained in the closure of their sum. Hence, $\sum_{j=1}^n \bar{Y}_j \subset \bar{Y}$. Since Y has been assumed to be closed, we have $\sum_{j=1}^n \bar{Y}_j \subset Y$.

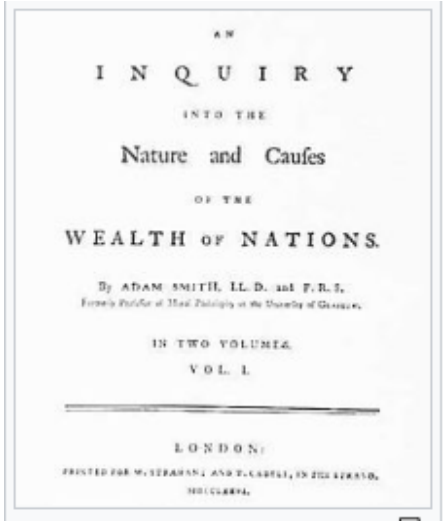
1. Introduction
2. The Simultaneous Optimization Approach
3. The Excess Demand Approach
4. Economies with A Measure Space of Agents

Introduction of General Equilibrium

1. Adam Smith (1776), **The Wealth of Nations**
   Invisible Hand = Equilibrium Price
2. Leon Walras (1874-1877), **Elements of Pure Economics**
   Walrasian Equilibrium
3. Kenneth J. Arrow and Gerard Debreu (1954), *Existence of an Equilibrium for a Competitive Economy*, Econometrica 22, 265 - 290
4. Gerard Debreu (1959), **Theory of Value – An Axiomatic Analysis of Economic Equilibrium**

Introduction





AN

INQUIRY

INTO THE

Nature and Causes

OF THE

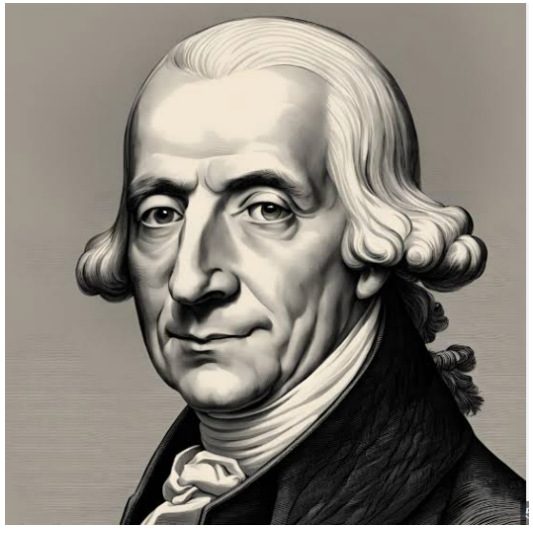
WEALTH OF NATIONS.

By ADAM SMITH, LL.D. and F.R.S.

IN TWO VOLUMES.

VOL. I.

LONDON:



Adam Smith (1776), **The Wealth of Nations**

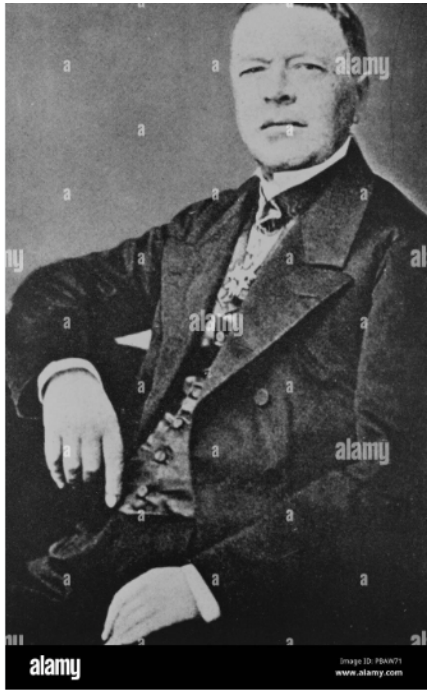
June 5 (OS) / June 16 (NS) 1723 - July 17, 1790

1. Scottish moral philosopher
2. Pioneering political economist

1. The Theory of Moral Sentiments (1759)
2. An Inquiry into the Nature and Causes of the Wealth of Nations (1776)

Introduction





Antoine Augustin Cournot (1838), **The Recherches**

August 28, 1801 - March 31, 1877

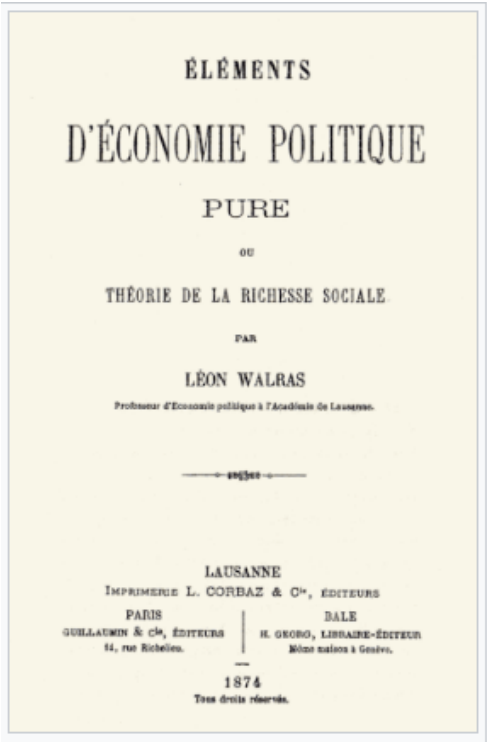
French economist, philosopher and mathematician

Researches into the Mathematical Principles of the Theory of Wealth (1838)

Introduction



Walras in 1862 (age 27)



ÉLÉMENTS

D'ÉCONOMIE POLITIQUE

PURE

OU

THÉORIE DE LA RICHESSE SOCIALE

PAR

LÉON WALRAS

Professeur d'Économie politique à l'Académie de Lausanne.

LAUSANNE

IMPRIMERIE L. CORBAZ & Cie, ÉDITEURS

PARIS

GUILLAUMIN & Cie, ÉDITEURS

14, rue Richelieu.

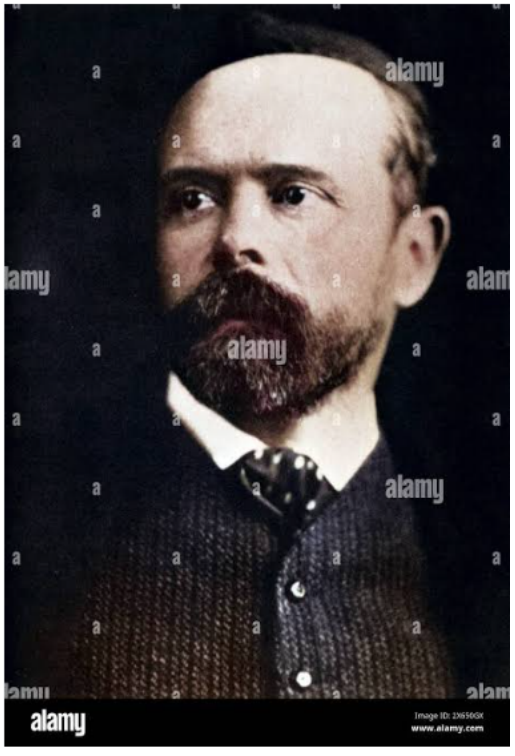
BALE

H. GEORG, LIBRAIRE-ÉDITEUR

Même maison à Genève.

1874

Tous droits réservés.



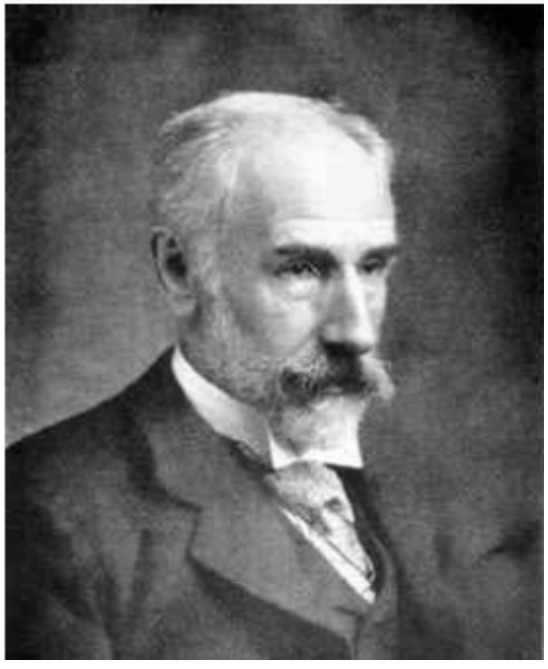
Marie-Esprit-Leon Walras (1874-1877),

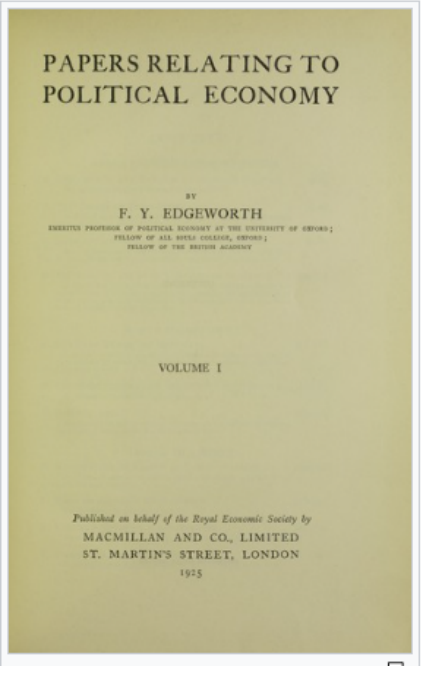
Elements of Pure Economics

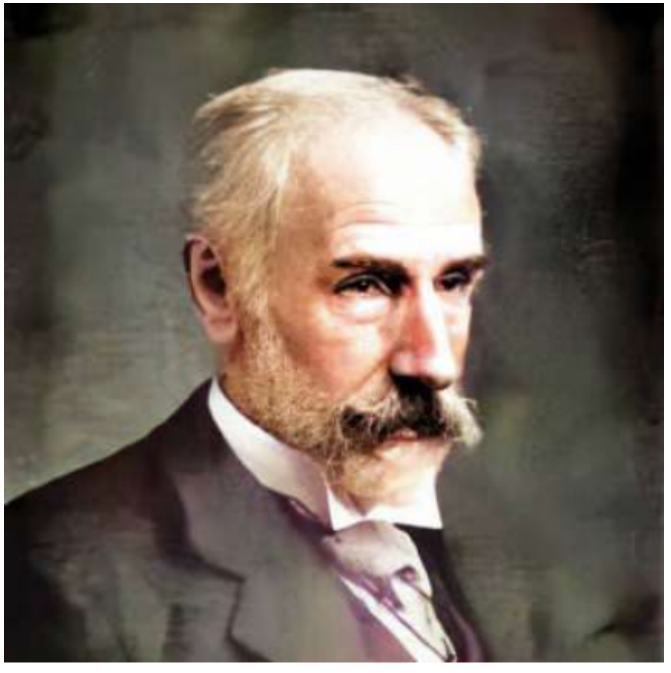
December 16, 1834 - January 5, 1910

Mathematical economist associated with the creation of the general equilibrium theory

Introduction



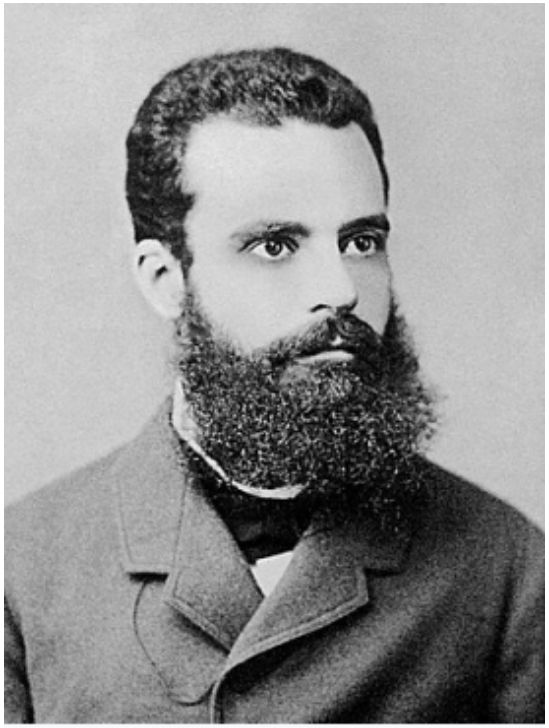




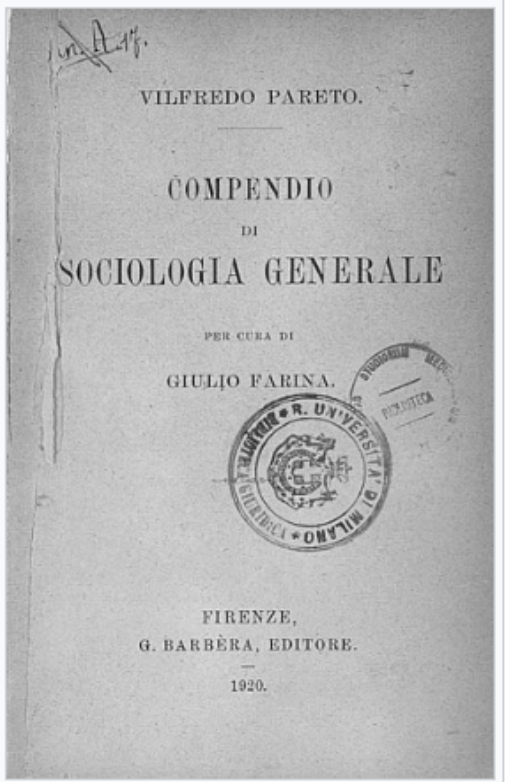
Francis Ysidro Edgeworth (1881), **Mathematical Physics**

- February 8, 1845 - February 13, 1926
- Mathematical Physics: An Essay on the Application of Mathematics to the Moral Sciences (1881)
- The Economic Journal (1891)
- Edgeworth's Limit Theorem

Introduction



Pareto in the 1870s





Vilfredo Federico Damaso Pareto (1896)

- July 15, 1848 - August 19, 1923
- French-Italian sociologist, economist and philosopher
- Pareto Efficiency / Pareto Optimality / Pareto Improvement

Introduction


von Neumann in the 1940s





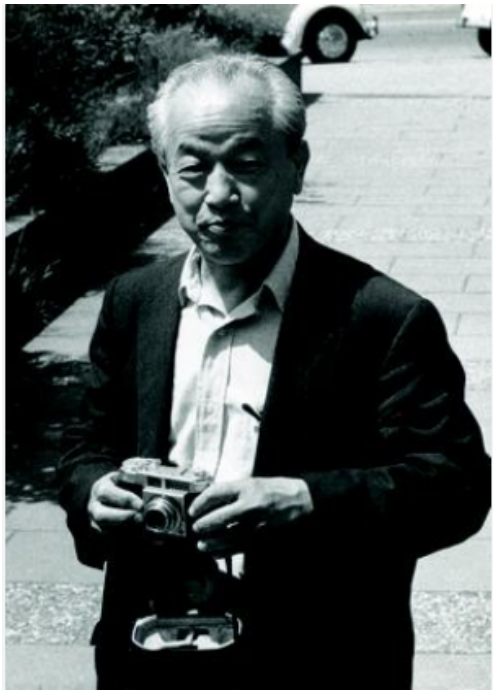
John von Neumann (1928, 1944)

- December 28, 1903 - February 8, 1957
- Mathematician, Economist, Computer Scientist, and so on
- Minimax Theorem (1928)
- Theory of Games and Economic Behavior (1944)

Introduction



Kakutani in 1970



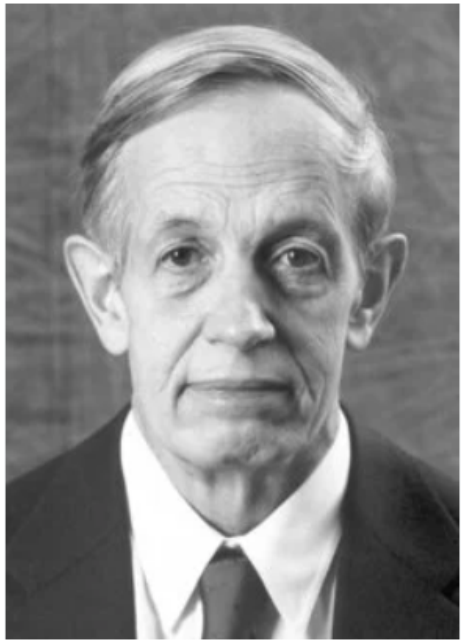


Shizuo Kakutani (1941)

- August 28, 1911 - August 17, 2004
- Japanese and American mathematician
- Kakutani's Fixed-Point Theorem (1941)

Introduction





John Forbes Nash Jr (1950)

- June 13, 1928 - May 23, 2015
- American mathematician
- Nash Equilibrium (1950)
- Kuhn - Tucker

Introduction

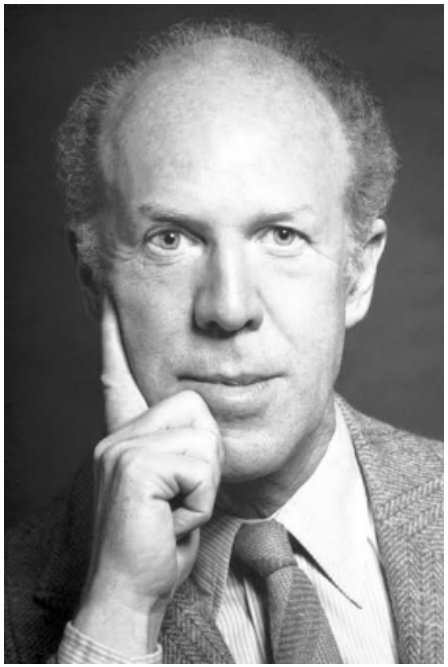




Kenneth Joseph Arrow (1950s)

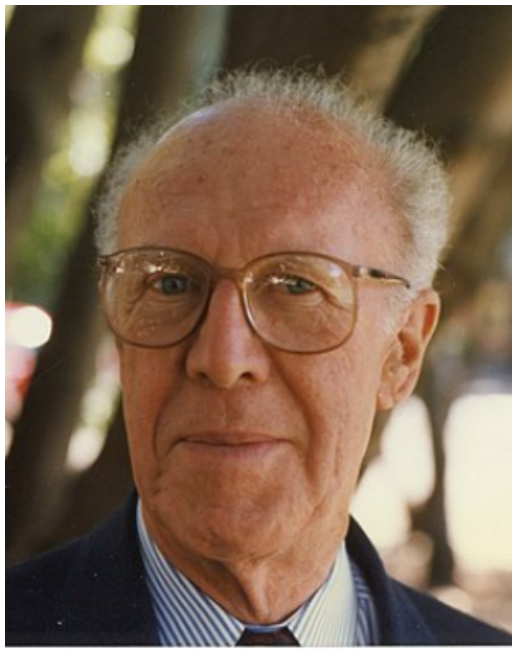
- August 23, 1921 - February 17, 2017
- American economist, mathematician and political throeist
- Social Choice Theory (Arrow's Impossibility Theorem)
- General Equilibrium Theory
- Fundamental Theorems of Welfare Economics
- Endogenous-Growth Theory
- Information Economics

Introduction





Debreu in 1977



Gerard Debreu (1954)

- July 4, 1921 - December 31, 2004
- French-born economist and mathematician
- General Equilibrium Theory (1954)

Kenneth J. Arrow and Gerard Debreu (1954), *Existence of an Equilibrium for a Competitive Economy*, Econometrica 22, 265 - 290

Mathematical Economics

1. Adam Smith: June 5 (OS) / June 16 (NS) 1723 - July 17, 1790
2. Antoine Augustin Cournot: August 28, 1801 - March 31, 1877
3. Marie-Esprit-Leon Walras: December 16, 1834 - January 5, 1910
4. Francis Ysidro Edgeworth: February 8, 1845 - February 13, 1926
5. Vilfredo Federico Damaso Pareto: July 15, 1848 - August 19, 1923
6. John von Neumann: December 28, 1903 - February 8, 1957
7. Shizuo Kakutani: August 28, 1911 - August 17, 2004
8. John Forbes Nash Jr: June 13, 1928 - May 23, 2015
9. Kenneth Joseph Arrow: August 23, 1921 - February 17, 2017
10. Gerard Debreu: July 4, 1921 - December 31, 2004

10 Economists Chosen by MIT Department of Economics

1. Adam Smith: June 5 (OS) / June 16 (NS) 1723 - July 17, 1790
2. David Richard: April 18, 1772 - September 11, 1823
3. Antoine Augustin Cournot: August 28, 1801 - March 31, 1877
4. John Stuart Mill: May 20, 1806 - May 8, 1873
5. Karl Heinrich Marx: May 5, 1818 - March 14, 1883
6. Marie-Esprit-Leon Walras: December 16, 1834 - January 5, 1910
7. Alfred Marshall: July 26, 1842 - July 13, 1924
8. John Maynard Keynes: June 5, 1883 - April 21, 1946
9. John von Neumann: December 28, 1903 - February 8, 1957
10. Joan Violet Robinson: October 31, 1903 - August 5, 1983

2 Great Economists in Century 20



1. Paul Anthony Samuelson: May 15, 1915 - December 13, 2009
2. Kenneth Joseph Arrow: August 23, 1921 - February 17, 2017

Introduction

There are FOUR distinct, but closely related, approaches to the existence problem of general equilibrium.

1. Fixed-Point Theorem [Brouwer (1911) and Kakutani (1941) and Analogous Arguments]
2. Efficient Algorithms — Approximate Economic Equilibrium [Scarf (1967)]
3. Fixed-Point Index + Degree Theory [Dierker (1972, 1974); Mityagin (1972); Smale (1974); Balasko (1975, 1978); Varian (1975); Nishimura (1978); Kalman and Lin (1978); and Kalman, Lin and Wiesmeth (1978)]
4. Differential Process — Generic Convergence [Smale (1976)]